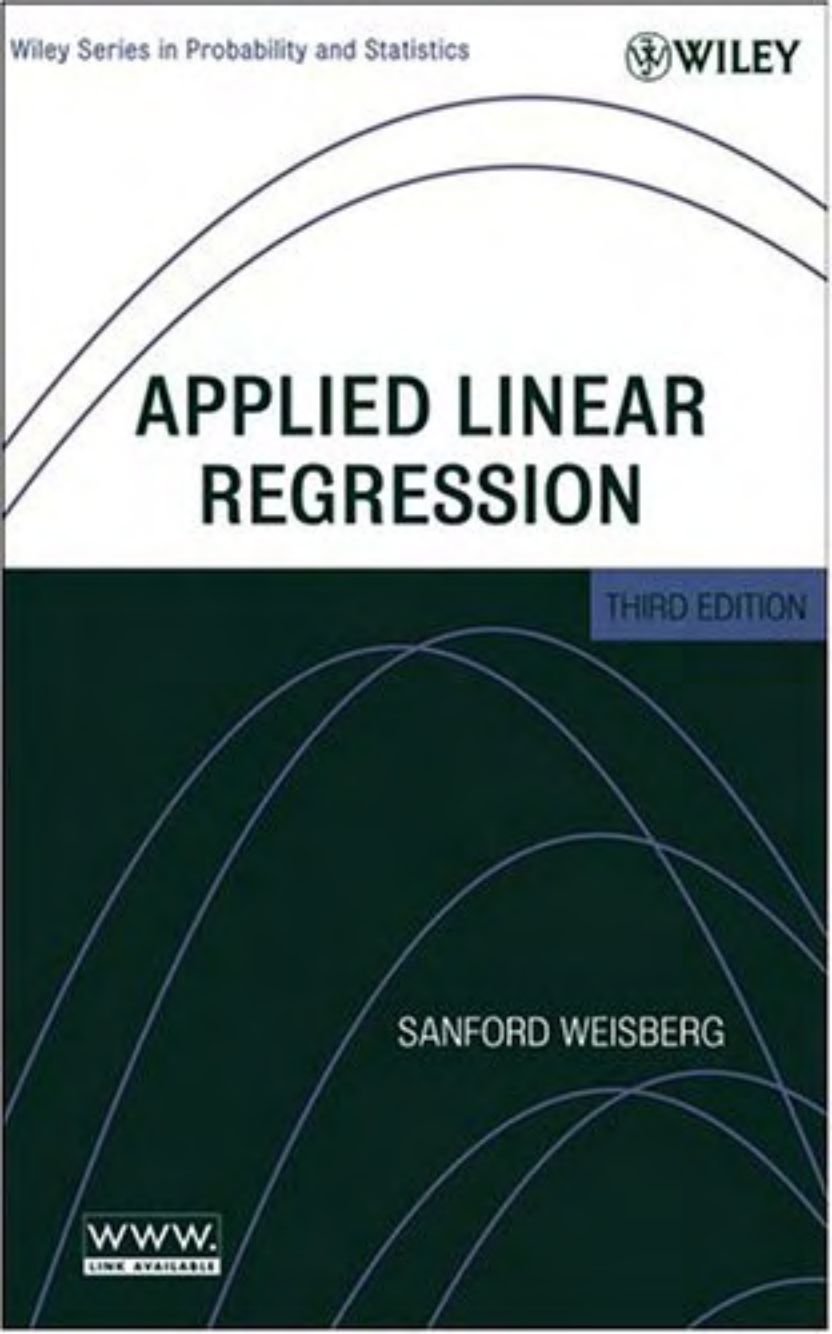
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APPLIED LINEAR REGRESSION

THIRD EDITION

SANFORD WEISBERG



Applied Linear Regression

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School of Statistics
Minneapolis, Minnesota



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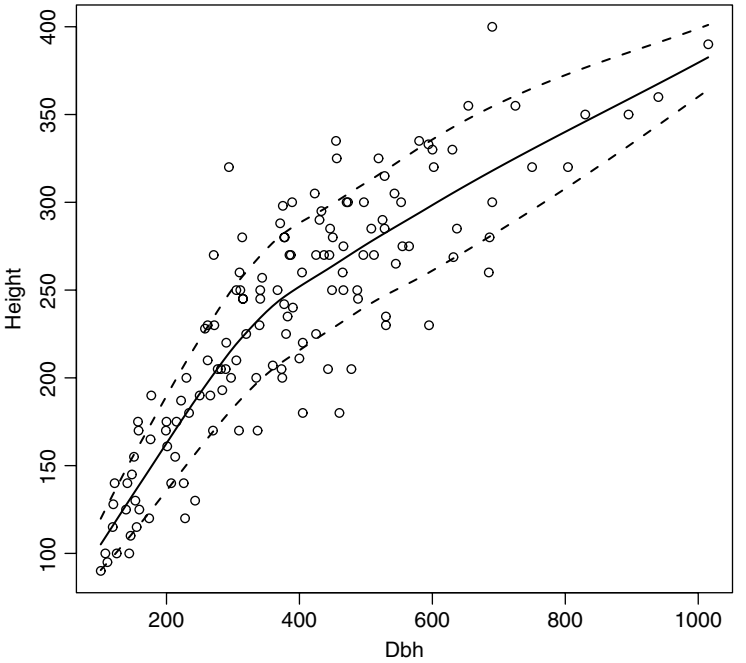


FIG. A.2 *loess* smooth for the mean function, solid line, and mean ± one standard deviation, dashed lines.

Figure A.2 shows the *loess* smooth for the mean function and the mean plus and minus one standard deviation for the same data as in Figure A.1. The variability appears to be a bit larger in the middle of the range than at the edges.

A.6 A BRIEF INTRODUCTION TO MATRICES AND VECTORS

We provide only a brief introduction to matrices and vectors. More complete references include Graybill (1969), Searle (1982), Schott (1996), or any good linear algebra book.

Boldface type is used to indicate matrices and vectors. We will say that $\mathbf{X}$ is an $r \times c$ matrix if it is an array of numbers with r rows and c columns. A specific 4×3 matrix $\mathbf{X}$ is

$$\mathbf{X} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 5 \\ 1 & 3 & 4 \\ 1 & 8 & 6 \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \\ x_{41} & x_{42} & x_{43} \end{pmatrix} = \left(x_{ij}\right) \tag{A.12}$$

The element x_{ij} of $\mathbf{X}$ is the number in the ith row and the jth column. For example, in the preceding matrix, $x_{32} = 3$.

A *vector* is a matrix with just one column. A specific 4×1 matrix $\mathbf{y}$, which is a vector of length 4, is given by

$$\mathbf{y} = \begin{pmatrix} 2 \\ 3 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

The elements of a vector are generally singly subscripted; thus, $y_3 = -2$. A *row vector* is a matrix with one row. We do not use row vectors in this book. If a vector is needed to represent a row, a transpose of a column vector will be used (see below).

A *square matrix* has the same number of rows and columns, so $r = c$. A square matrix $\mathbf{Z}$ is *symmetric* if $z_{ij} = z_{ji}$ for all i and j. A square matrix is *diagonal* if all elements off the main diagonal are 0, $z_{ij} = 0$, unless $i = j$. The matrices $\mathbf{C}$ and $\mathbf{D}$ below are symmetric and diagonal, respectively:

$$\mathbf{C} = \begin{pmatrix} 7 & 3 & 2 & 1 \\ 3 & 4 & 1 & -1 \\ 2 & 1 & 6 & 3 \\ 1 & -1 & 3 & 8 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 7 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

The diagonal matrix with all elements on the diagonal equal to 1 is called the *identity matrix*, for which the symbol $\mathbf{I}$ is used. The 4×4 identity matrix is

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A *scalar* is a 1×1 matrix, an ordinary number.

A.6.1 Addition and Subtraction

Two matrices can be added or subtracted only if they have the same number of rows and columns. The sum $\mathbf{C} = \mathbf{A} + \mathbf{B}$ of $r \times c$ matrices is also $r \times c$. Addition is done elementwise:

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \\ a_{31} + b_{31} & a_{32} + b_{32} \end{pmatrix}$$

Subtraction works the same way, with the "+" signs changed to "−" signs. The usual rules for addition of numbers apply to addition of matrices, namely commutativity, $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$, and associativity, $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$.

A.6.2 Multiplication by a Scalar

If k is a number and $\mathbf{A}$ is an $r \times c$ matrix with elements (a_{ij}), then $k\mathbf{A}$ is an $r \times c$ matrix with elements (ka_{ij}). For example, the matrix $\sigma^2\mathbf{I}$ has all diagonal elements equal to σ^2 and all off-diagonal elements equal to 0.

A.6.3 Matrix Multiplication

Multiplication of matrices follows rules that are more complicated than are the rules for addition and subtraction. For two matrices to be multiplied together in the order $\mathbf{AB}$, the number of columns of $\mathbf{A}$ must equal the number of rows of $\mathbf{B}$. For example, if $\mathbf{A}$ is $r \times c$, and $\mathbf{B}$ is $c \times q$, then $\mathbf{C} = \mathbf{AB}$ is $r \times q$. If the elements of $\mathbf{A}$ are (a_{ij}) and the elements of $\mathbf{B}$ are (b_{ij}), then the elements of $\mathbf{C} = (c_{ij})$ are given by the formula

$$c_{ij} = \sum_{k=1}^{c} a_{ik} b_{kj}$$

This formula says that c_{ij} is formed by taking the ith row of $\mathbf{A}$ and the jth column of $\mathbf{B}$, multiplying the first element of the specified row in $\mathbf{A}$ by the first element in the specified column in $\mathbf{B}$, multiplying second elements, and so on, and then adding the products together.

If $\mathbf{A}$ is $1 \times c$ and $\mathbf{B}$ is $c \times 1$, then the product $\mathbf{AB}$ is 1×1, an ordinary number. For example, if $\mathbf{A}$ and $\mathbf{B}$ are

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 2 & -1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 2 \\ 1 \\ -2 \\ 4 \end{pmatrix}$$

then the product $\mathbf{AB}$ is

$$\mathbf{AB} = (1 \times 2) + (3 \times 1) + (2 \times -2) + (-1 \times 4) = -3$$

$\mathbf{AB}$ is not the same as $\mathbf{BA}$. For the preceding matrices, the product $\mathbf{BA}$ will be a 4×4 matrix:

$$\mathbf{BA} = \begin{pmatrix} 2 & 6 & 4 & -2 \\ 1 & 3 & 2 & -1 \\ -2 & -6 & -4 & 2 \\ 4 & 12 & 8 & -4 \end{pmatrix}$$

The following small example illustrates what happens when all the dimensions are bigger than 1. A 3×2 matrix $\mathbf{A}$ times a 2×2 matrix $\mathbf{B}$ is given as

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} \end{pmatrix}$$

Using numbers, an example of multiplication of two matrices is

$$\begin{pmatrix} 3 & 1 \\ -1 & 0 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 15+0 & 3+4 \\ -5+0 & -1+0 \\ 10+0 & 2+8 \end{pmatrix} = \begin{pmatrix} 15 & 4 \\ -5 & -1 \\ 10 & 10 \end{pmatrix}$$

In this example, **BA** is not defined because the number of columns of **B** is not equal to the number of rows of **A**. However, the associative law holds: If **A** is $r \times c$, **B** is $c \times q$, and **C** is $q \times p$, then $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$, and the result is an $r \times p$ matrix.

A.6.4 Transpose of a Matrix

The transpose of an $r \times c$ matrix **X** is a $c \times r$ matrix called $\mathbf{X}'$ such that if the elements of **X** are (x_{ij}), then the elements of $\mathbf{X}'$ are (x_{ji}). For the matrix **X** given at (A.12),

$$\mathbf{X}' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 3 & 8 \\ 1 & 5 & 4 & 6 \end{pmatrix}$$

The transpose of a column vector is a row vector. The transpose of a product $(\mathbf{AB})'$ is the product of the transposes, in *opposite order*, so $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$.

Suppose that **a** is an $r \times 1$ vector with elements $a_1, \ldots, a_r$. Then the product $\mathbf{a}'\mathbf{a}$ will be a 1×1 matrix or scalar, given by

$$\mathbf{a}'\mathbf{a} = a_1^2 + a_2^2 + \cdots + a_r^2 = \sum_{i=1}^{r} a_i^2 \tag{A.13}$$

Thus, $\mathbf{a}'\mathbf{a}$ provides a compact notation for the sum of the squares of the elements of a vector **a**. The square root of this quantity $(\mathbf{a}'\mathbf{a})^{1/2}$ is called the *norm* or *length* of the vector **a**. Similarly, if **a** and **b** are both $r \times 1$ vectors, then we obtain

$$\mathbf{a}'\mathbf{b} = a_1b_1 + a_2b_2 + \cdots + a_nb_n = \sum_{i=1}^{r} a_ib_i = \sum_{i=1}^{r} b_ia_i = \mathbf{b}'\mathbf{a}$$

The fact that $\mathbf{a}'\mathbf{b} = \mathbf{b}'\mathbf{a}$ is often quite useful in manipulating the vectors used in regression calculations.

Another useful formula in regression calculations is obtained by applying the distributive law

$$(\mathbf{a} - \mathbf{b})'(\mathbf{a} - \mathbf{b}) = \mathbf{a}'\mathbf{a} + \mathbf{b}'\mathbf{b} - 2\mathbf{a}'\mathbf{b} \tag{A.14}$$

A.6.5 Inverse of a Matrix

For any scalar $c \neq 0$, there is another number called the *inverse* of c, say d, such that the product $cd = 1$. For example, if $c = 3$, then $d = 1/c = 1/3$, and the inverse

of 3 is $1/3$. Similarly, the inverse of $1/3$ is 3. The number 0 does not have an inverse because there is no other number d such that $0 \times d = 1$.

Square matrices can also have an inverse. We will say that the inverse of a matrix $\mathbf{C}$ is another matrix $\mathbf{D}$, such that $\mathbf{CD} = \mathbf{I}$, and we write $\mathbf{D} = \mathbf{C}^{-1}$. Not all square matrices have an inverse. The collection of matrices that have an inverse are called *full rank*, *invertible*, or *nonsingular*. A square matrix that is not invertible is of less than full rank, or *singular*. If a matrix has an inverse, it has a unique inverse.

The inverse is easy to compute only in special cases, and its computation in general can require a very tedious calculation that is best done on a computer. High-level matrix and statistical languages such as Matlab, Maple, Mathematica, R and S-plus include functions for inverting matrices, or returning an appropriate message if the inverse does not exist.

The identity matrix $\mathbf{I}$ is its own inverse. If $\mathbf{C}$ is a diagonal matrix, say

$$\mathbf{C} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

then $\mathbf{C}^{-1}$ is the diagonal matrix

$$\mathbf{C} = \begin{pmatrix} \frac{1}{3} & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

as can be verified by direct multiplication. For any diagonal matrix with nonzero diagonal elements, the inverse is obtained by inverting the diagonal elements. If any of the diagonal elements are 0, then no inverse exists.

A.6.6 Orthogonality

Two vectors $\mathbf{a}$ and $\mathbf{b}$ of the same length are *orthogonal* if $\mathbf{a}'\mathbf{b} = 0$. An $r \times c$ matrix $\mathbf{Q}$ has *orthonormal columns* if its columns, viewed as a set of $c \leq r$ different $r \times 1$ vectors, are orthogonal and in addition have length 1. This is equivalent to requiring that $\mathbf{Q}'\mathbf{Q} = \mathbf{I}$, the $r \times r$ identity matrix. A square matrix $\mathbf{A}$ is *orthogonal* if $\mathbf{A}'\mathbf{A} = \mathbf{A}\mathbf{A}' = \mathbf{I}$, and so $\mathbf{A}^{-1} = \mathbf{A}'$. For example, the matrix

$$\mathbf{A} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix}$$

can be shown to be orthogonal by showing that $\mathbf{A}'\mathbf{A} = \mathbf{I}$, and therefore

$$\mathbf{A}^{-1} = \mathbf{A}' = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}$$

A.6.7 Linear Dependence and Rank of a Matrix

Suppose we have a $n \times p$ matrix $\mathbf{X}$, with columns given by the vectors $\mathbf{x}_1, \ldots, \mathbf{x}_p$; we consider only the case $p \leq n$. We will say that $\mathbf{x}_1, \ldots, \mathbf{x}_p$ are *linearly dependent* if we can find multipliers $a_1, \ldots, a_p$, not all of which are 0, such that

$$\sum_{i=1}^{p} a_i \mathbf{x}_i = \mathbf{0} \tag{A.15}$$

If no such multipliers exist, then we say that the vectors are *linearly independent*, and the matrix is *full rank*. In general, the *rank* of a matrix is the maximum number of $\mathbf{x}_i$ that form a linearly independent set.

For example, the matrix $\mathbf{X}$ given at (A.12) can be shown to have linearly independent columns because no a_i not all equal to zero can be found that satisfy (A.15). On the other hand, the matrix

$$\mathbf{X} = \begin{pmatrix} 1 & 2 & 5 \\ 1 & 1 & 4 \\ 1 & 3 & 6 \\ 1 & 8 & 11 \end{pmatrix} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \tag{A.16}$$

has linearly dependent columns and is singular because $\mathbf{x}_3 = 3\mathbf{x}_1 + \mathbf{x}_2$, or $3\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_3 = \mathbf{0}$. This matrix is of rank two because the linearly independent subset of the columns with the most elements, consisting of any two of the three columns, has two elements.

The matrix $\mathbf{X}'\mathbf{X}$ is a $p \times p$ matrix. If $\mathbf{X}$ has rank p, so does $\mathbf{X}'\mathbf{X}$. Full-rank square matrices always have an inverse. Square matrices of less than full rank never have an inverse.

A.7 RANDOM VECTORS

An $n \times 1$ vector $\mathbf{Y}$ is a *random vector* if each of its elements is a random variable. The mean of an $n \times 1$ random vector $\mathbf{Y}$ is also an $n \times 1$ vector whose elements are the means of the elements of $\mathbf{Y}$. The variance of an $n \times 1$ vector $\mathbf{Y}$ is an $n \times n$ square symmetric matrix, often called a *covariance matrix*, written $\mathrm{Var}(\mathbf{Y})$ with $\mathrm{Var}(y_i)$ as its (i, i) element and $\mathrm{Cov}(y_i, y_j) = \mathrm{Cov}(y_j, y_i)$ as both the (i, j) and (j, i) element.

The rules for means and variances of random vectors are matrix equivalents of the scalar versions in Appendix A.2. If $\mathbf{a}_0$ is a vector of constants, and $\mathbf{A}$ is a matrix of constants,

$$\mathrm{E}(\mathbf{a}_0 + \mathbf{AY}) = \mathbf{a}_0 + \mathbf{A}\mathrm{E}(\mathbf{Y}) \tag{A.17}$$

$$\mathrm{Var}(\mathbf{a}_0 + \mathbf{AY}) = \mathbf{A}\mathrm{Var}(\mathbf{Y})\mathbf{A}' \tag{A.18}$$

A.8 LEAST SQUARES USING MATRICES

The multiple linear regression model can be written as

$$\mathrm{E}(Y|X = \mathbf{x}) = \boldsymbol{\beta}'\mathbf{x}$$

$$\mathrm{Var}(Y|X = \mathbf{x}) = \sigma^2$$

In matrix terms, we will write the model using errors as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

where $\mathbf{Y}$ is the $n \times 1$ vector of response values and $\mathbf{X}$ is a $n \times p'$ matrix. If the mean function includes an intercept, then the first column of $\mathbf{X}$ is a vector of ones, and $p' = p + 1$. If the mean function does not include an intercept, then the column of one is not included in $\mathbf{X}$ and $p' = p$. The ith row of the $n \times p'$ matrix $\mathbf{X}$ is $\mathbf{x}_i'$, $\boldsymbol{\beta}$ is a $p' \times 1$ vector of parameters for the mean function, $\mathbf{e}$ is the $n \times 1$ vector of unobservable errors, and σ^2 is an unknown positive constant.

The OLS estimate $\hat{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}$ is given by the arguments that minimize the residual sum of squares function,

$$RSS(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

Using (A.14), we obtain

$$RSS(\boldsymbol{\beta}) = \mathbf{Y}'\mathbf{Y} + \boldsymbol{\beta}'(\mathbf{X}'\mathbf{X})\boldsymbol{\beta} - 2\mathbf{Y}'\mathbf{X}\boldsymbol{\beta} \tag{A.19}$$

$RSS(\boldsymbol{\beta})$ depends on only three functions of the data: $\mathbf{Y}'\mathbf{Y}$, $\mathbf{X}'\mathbf{X}$, and $\mathbf{Y}'\mathbf{X}$. Any two data sets that have the same values of these three quantities will have the same least squares estimates. Using (A.8), the information in these quantities is equivalent to the information contained in the sample means of the terms plus the sample covariances of the terms and the response.

To minimize (A.19), differentiate with respect to $\boldsymbol{\beta}$ and set the result equal to 0. This leads to the matrix version of the *normal equations*,

$$\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{Y} \tag{A.20}$$

The OLS estimates are any solution to these equations. If the inverse of $(\mathbf{X}'\mathbf{X})$ exists, as it will if the columns of $\mathbf{X}$ are linearly independent, the OLS estimates are unique and are given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \tag{A.21}$$